

Equilibrium Tape Synth (ETS)

A frozen world, a causal writer, and one free energy

ETS Project

(build documentation; author line placeholder)

Post-v1 documentation

Abstract

ETS learns a *frozen “world”* from N real music tracks and lets a performer *play* a causal writer through *six control lanes*. The world is not a sample generator: it is a single free-energy functional F whose equilibria are the real tracks, fit by contrastive estimation against the tracks’ own disarrangements. The output is a *settled tape* — a progressively clamped $(N+1)$ -th boundary node of the same hypergraph as the input tracks — whose every emitted sample is a real source unit at a real metrical slot, provenance-traceable to `(track, unit, transform)`. Control never edits the world; it enters as a single *exponential tilt* on gauge-invariant arrangement statistics — the Doob conditioning induced by clamping the panel as a boundary measure. The system’s distinctive commitment is its *method*: a two-state invariant manifest (14 of 15 invariants enforced by executable checks), an adversarial builder/auditor pair, prereg-commit-before-run, and an append-only registry in which *walls are information* and *silent divergence is the breach class*. This paper leads with the architecture, gives the Gibbs energy-based model its due as the centerpiece, and states plainly what is real versus pending — including a headline dissociation gate (G2, holonomy above null) that *did not pass* at v0 and is reported as-is rather than retuned.

Authorities. `ets-spec-v0.md` (functional §5, invariants §14, panel §8, render §11); `ets-connector-v0.md` (Layer-0 tilt map); `REGISTRY.json1` (build history, honest numbers). Where those and this paper disagree, they win. Every figure below is traced to a committed source or results artifact.

1 Architecture overview (read this first)

ETS is a pipeline from real audio to a settled tape. Each stage has a plain-terms job and a formal object; the design’s whole point is that *exactly one stage makes value judgments* — the functional F — and every other stage measures, transports, applies, or reports. Figure 1 is the labelled flow.

There are two orthogonal control paths, and only two: (i) the six **CV lanes** enter the writer through a *single jack*, the h-transform tilt (I-1); (ii) **clamps** — committed tape, “this unit at bar 33”, and knob leans as clamped role-space mass — are all the same boundary-condition species (I-7, generalized by the connector). No third intervention mechanism exists. The desktop runtime is *two processes* (§6): an **ENGINE** (Python core, streaming writer, audio) and a **PANEL** (PySide6/Qt) over OSC on localhost; killing the panel never touches the engine, and any OSC/MIDI hardware can replace the panel.

2 Ingestion, gauge, and the frozen world

Ingestion. Audio is resampled to 44.1 kHz, beat-tracked (`beat_this==1.1.0`) to a tatum/beat/bar grid, and decomposed by a fixed 8-band log-spaced filterbank — deliberately *not* a separator, because “a separator is a second authority pre-deciding roles before F does” (spec §2.3). The raised-cosine masks form a *partition of unity* (`ingestion/filterbank.py`: `N_FFT=2048`, `HOP=512`, `FMIN=40 Hz`);

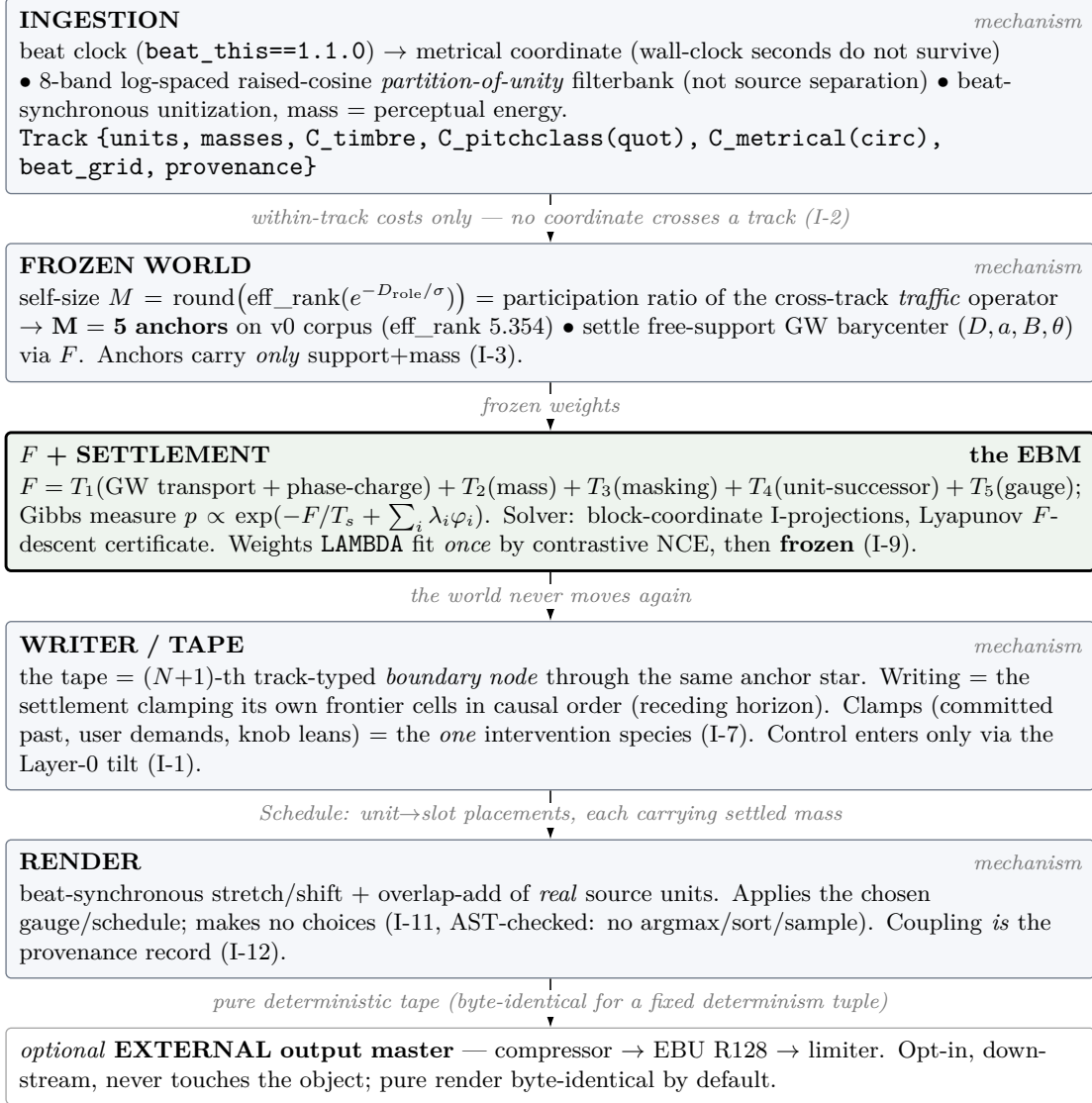


Figure 1: The ETS pipeline. The energy-based model (bold box) is the only stage that makes value judgments; every other stage measures, transports, applies, or reports. Invariant tags (I- k) mark where each faithfulness law binds.

the per-band ISTFTs sum back to the input to numerical precision, which makes the G0 reconstruction identity a coverage/scheduling test rather than a lossy-model test.

Gauge (spec §3). Per track the gauge group is transposition $\times$ beat-phase $\times$ loudness $\times$ timbre-basis. All cross-track communication is invariant under independent per-track gauge action; *no coordinate crosses a track boundary* (I-2: descriptor arrays are private, the sole combiner refuses a cross-track pair, pitch-class cost is transposition-quotiented, metrical cost is circular). Microtiming deviation is *not* gauge — it is intrinsic groove content that must survive, and it is exactly what T_1 ’s phase charge later prices.

Frozen world (self-sizing anchors, spec §4). All cross-track traffic factors through free-support barycenter *anchors* in role space; there is no direct pairwise track coupling. The anchor count is not a hyperparameter — it is *derived* as the balanced-truncation effective rank (participation ratio $(\sum w)^2 / \sum w^2$) of the cross-track GW role-affinity operator $e^{-D_{\text{role}}/\sigma}$ (`anchors.py:effective_rank`). On the v0 world this is $M = 5$ ($\text{eff_rank} \approx 5.354$, $\sigma \approx 0.553$). The supports (D, a, B, θ) are then settled as the GW barycenter at that count. Anchors hold *only* support and mass — no accumulator, EMA, or momentum (I-3, structurally scanned).

Honest note. The effective-rank rule is a *re-derivation after a wall*: the first sizing rule (a GW transport-residual growth criterion) provably failed to dissociate role-diverse from same-role corpora and grew with N , and was rejected; the current rule reads the spec’s own “McMillan degree of *traffic* / Hankel mass / balanced truncation” language as a spectral rank (`RETRO_AUDIT.md`). The wall was reported and the object re-posed, not patched.

3 The Gibbs energy-based model (centerpiece)

The heart of ETS is a *single free-energy functional* F (invariant I-4: there is no training loss distinct from F). Framed correctly, ETS is an **energy-based model**: the energy is F , the sampling looseness is a *temperature* T_s , and control is an *exponential tilt* in sufficient statistics φ . Real tracks are the low-energy equilibria; their rearrangements are not.

3.1 F over the unit-resolved coupling

F ranges over couplings π (unit $\rightarrow$ role $\rightarrow$ metrical slot), channel gains B , anchor supports/masses (D, a) , and gauge sections g . Crucially, F is posed on the *full unit-resolved* π , not on its role-occupancy marginal $O[k, s] = \sum_{\text{units}} \pi$. A term may use the marginal *only where it provably factors through it*; a term that needs the fiber (which real unit sits where) reads π directly (Table 1).

Table 1: The five terms of F (`functional/f.py`, rev-r1).

Term	Meaning	Posed on	Reads
T_1	GW transport <i>plus</i> a circular metrical phase-displacement charge	full- π	GW via prototypes; charge via th
T_2	mass conservation (unbalanced-OT gen.-KL of O vs $a\theta$)	marginal O	with a written factorization proof
T_3	spectral masking (collision $\sum E^2$ – self, $E = O^\top B$)	marginal O	with a written factorization proof
T_4	unit-successor run-continuation (Doob h -transform)	full- π	π ’s fiber — inexpressible over O
T_5	per- <i>section</i> gauge-fixing (global transpose/phase), never per-unit	gauge	neither π nor O

The **phase-displacement charge** is what makes groove cost something. For each placed unit it charges the circular distance between the unit’s *intrinsic* metrical coordinate x_u and the phase

of the slot it occupies, quotiented by a single per-section global phase shift δ (a global beat-phase shift is free gauge; a per-unit displacement is charged). The δ -quotient is solved in closed form,

$$\min_{\delta} \sum_u m_u (1 - \cos 2\pi(x_u - \delta)) = 1 - \left| \sum_u m_u e^{i2\pi x_u} \right| \quad (\text{masses normalized}), \quad (1)$$

which is exactly 0 when every unit sits at its own intrinsic slot up to one global shift (a real track’s groove) and strictly positive for incoherent displacement. T_4 rewards output-adjacent slots holding genuine *source* successors: a grid-shuffle re-deals content so almost no adjacency survives; a cross-track graft inserts units that are no track’s successor, so the run breaks. Both fiber terms are gauge-invariant by construction (I-2). Holonomy, drift, novelty, and any meter appear *nowhere* in F (I-5, I-14; enforced by an AST scan that bites on a `holonomy_drift` identifier and on any `ets.meters` import in the F path).

3.2 The Gibbs measure and the Layer-0 tilt

The equilibrium measure over an arrangement a , with control applied, is

$$p(a) \propto \exp\left(-\frac{F(a)}{T_s} + \sum_i \lambda_i \varphi_i(a)\right), \quad \lambda_i = \frac{u_i}{\sigma_{\varphi_i}} \quad (\text{connector Layer 0}). \quad (2)$$

Here T_s is the *temperature* (sampling looseness around the settled optimum; it carries no φ); the φ_i are five gauge-invariant arrangement statistics (region, density, continuity, gauge-move magnitude, novelty) computable from the candidate arrangement alone (`connector/phi.py`). The tilt is not a bolt-on: it is the *Doob conditioning* the settlement inherits when the panel’s boundary measure is clamped. Two of the five φ (region, density) are linear in O and enter the O-block; continuity and novelty live on the fiber block; gauge on the (v0-frozen) gauge block. Run-time knobs move λ and T_s , never F ’s term weights (I-9).

3.3 Contrastive (NCE) training — and the honest KILL

The weights `LAMBDA` are fit *once*, at corpus time, by a convex logistic NCE (spec §6, `training/nce.py`). The condition is: *each real track is an equilibrium of F ; its rearrangements are not*. There is *no external “bad music” data anywhere* (I-6): the comparison class is generated internally by disarranging each real track’s own units, drawn only from a fixed, pre-registered scramble family (Table 2). Because F is linear in the weights and φ is computed at the frozen `LAMBDA`-free world, the NCE objective is *convex* with an exact gradient — no circularity. The fit uses seeds $\{1, 2, 3\}$; held-out validity is measured on *disjoint* seeds $\{4, 5\}$, so no fit metric is a gate metric (I-5).

Table 2: The fixed pre-registered scramble family (the only negatives).

Scramble	What it breaks	Arity
grid-shuffle	metrical placement (re-deal real units within a band)	Track→Track
phase-rotate	the single gauge-phase frame (incoherent <i>per-band</i> rotation)	Track→Track
role-permute	role assignment (derange anchor columns of the pure-GW coupling)	(Track, world)→Arr.
cross-track-swap	anchor-mediated cross-track coherence (swap anchor rows)	(Track ² , world)→Arr.

The KILL, then rev-r1 (load-bearing). The *first* F was posed on the occupancy *marginal* O . It **failed**: on the marginal, F could not separate real from grid-shuffle — held-out separation ≈ 0.35 , a *negative margin* — because the marginal is blind to *which* unit sits at *which* slot (registry `train-nce-2026-07-13 KILL`). This was ruled a *fidelity breach*: a silent aggregate projection of π , an unfaithful implementation *plus* a missing invariant (registry `fidelity-breach-2026-07-13`). Per the wall protocol it was reported, not patched: no LAMBDA was emitted, the referee was left incorruptible, and F was re-posed on the full unit-resolved fiber (rev-r1). A new invariant, **I-15** (no premature aggregation / structure-deleting projection), was added to forbid the class. The re-posed F (fork C: richer fiber + gauge-aligned groove target) **passed** (Table 3; `training_results.json`, registry `train-nce-revr1-2026-07-13`).

Table 3: Held-out per-member separation, rev-r1 fork C (disjoint seeds $\{4, 5\}$).

Family member	held-out separation	median margin
grid-shuffle	1.00	18.79
role-permute	0.95	1.07
phase-rotate	1.00	6.02
cross-track-swap	0.975	7.33
overall min	0.95 ≥ 0.90 (pre-registered)	—

(logistic loss 0.0908, grad-norm 0.0024, 240 pairs.) The emitted weights, now *frozen* and read live by the writer (I-9), are

$$\text{LAMBDA} = \{ T_2:4.9923, T_3:0.8023, T_4:10.4577, T_{1p}:8.7581, T_5:0.1 \},$$

with T_1 ’s GW transport as the reference scale (weight 1). $T_5 = 0.1$ is a run-time gauge baseline, *not* corpus-identifiable (a global section-gauge move is orthogonal to every rearrangement, so it carries no contrastive signal). An independent audit re-ran the fit and reproduced LAMBDA bit-exactly to 16 digits.

3.4 Settlement as certified I-projections

Minimization is block-coordinate I-projections (`functional/solver.py`): Sinkhorn on π -blocks, exponentiated-gradient (mirror descent) on B -blocks, unbalanced multiplicative updates on masses, closed-form GW barycenter on D , exact minimization on the gauge block. The certificate is a *Lyapunov F -descent guard*: every block step is accepted *only* if F does not increase, so the F -trajectory is monotone non-increasing and the run stops at $|\Delta F| < \text{tol}$. The accept/reject decision reads F and nothing else — no second objective, no meter (I-4/I-14, verified by parsing `batch_solve` and confirming the guard compares $F_{\text{cand}}/F_{\text{cur}}$). In streaming mode the batch Lyapunov certificate becomes a per-step frontier F -descent plus a *bounded-state* certificate: state growth on stationary input is a broken instrument — halt and report (I-8).

4 The control surface

Six CV lanes, exhaustive (spec §8); a seventh requires a spec revision (Table 4).

The knob-scaling law is derived, not hand-set. Each lane’s scale is $\lambda_i = u_i/\sigma_{\varphi_i}$, where σ_{φ_i} is the *equilibrium fluctuation* of φ_i under the *untitled* writer, measured by a registered calibration

Table 4: The six lanes, their φ statistics (`connector/phi.py`), and v0 status.

#	Lane	φ statistic	v0 status
1	REGION TILT (XY / channel strips)	φ_{region} : mass by anchor role	armed (σ per anchor)
2	DENSITY	φ_{density} : bar scheduled mass	disarmed ($\sigma=0$ at $u=0$)
3	CONTINUITY $\leftrightarrow$ RECOMB.	φ_{cont} : source-successor count	armed ($\sigma \approx 0.301$)
4	GAUGE STIFFNESS	φ_{gauge} : frame-move magnitude	disarmed ($\sigma=0$, frozen frame)
5	NOVELTY PRESSURE	φ_{novelty} : $1/\Delta$ recency reuse	armed ($\sigma \approx 0.330$)
6	TEMPERATURE	(<i>no</i> φ): settlement sharpness	armed

pass at world-freeze (`calibration/sigma_phi.json`). The derivation is fluctuation–dissipation: with $p_u \approx p_0 e^{\lambda\varphi}$ one has $d\langle\varphi\rangle/d\lambda|_0 = \text{Var}_0(\varphi)$, so $\lambda = u/\sigma$ gives $d\langle\varphi\rangle/du|_0 = \sigma$ — *one knob unit = one equilibrium- σ lean*. σ is the unique normalizer with this property and makes the tilt invariant to any rescaling of φ . No hand-set λ scale exists in the runtime. The calibration ensemble was 7495 bars (the corpus bar count — the evidence scale, not a hand constant), $s_{\text{phase}} = 8$.

Two lanes are honestly disarmed at v0. The untilted writer settles by a deterministic MAP descent on a bar-periodic anchor field, so φ_{density} (an O -marginal) is *pinned* constant ($\sigma = 0$ exactly), and the single identity-gauge section makes $\varphi_{\text{gauge}} \equiv 0$. Rather than invent a floor, the loader marks these lanes `identifiable=false`: u still transmits, but no tilt is applied and the engine surfaces the state (`WorldNotCalibrated` on a lean into an uncalibrated world). This is a theorem about degenerate exponential families reported as a wall, not an error-hiding branch.

Two processes over OSC (spec §12). The ENGINE (`ets/engine`) owns the frozen world, the streaming writer, and audio (`sounddevice`, `lazy`, `headless-graceful`). The PANEL (`ets/panel`, PySide6/Qt) is a pure control surface: six lanes, XY vector pad, meter jacks as LEDs, patch-cable feedback metaphor, MIDI CC learn. IPC is OSC over localhost (`python-osc`); no web technology anywhere in the runtime (I-13, enforced by an import scan that even distinguishes the runtime’s own `ets.panel` from the forbidden HoloViz `panel`). Control latency is a *declared* L bars measured by buffer math on the host (desktop: $L = 2$ bars, T_{prod} mean $0.568\text{s} < T_{\text{bar}} 1.486\text{s}$; `latency_desktop.json`); an underrun inside $L \cdot T_{\text{bar}}$ is a WALL — there is no degraded-quality fallback.

Meter jacks. Read-only, never in any objective/gradient/settlement (I-5). The conflated DRIFT jack was split into two shadow meters — `slide` (per-bar displacement of the settled gauge frame, in F ’s own quotient) and `loop` (a per-committed-bar transplant of the G2 loop-defect integrator) — then the old conflated jack was *deleted* after a registered one-shot regression showed it is exactly informationless on every producible trace (residual 0.0 at machine precision, registry `conflation-regression-stage1`).

5 The faithfulness discipline (the project’s identity)

ETS treats faithfulness as an engineering surface with teeth, not a virtue.

Two-state invariant manifest (`tests/invariants/manifest.py`). Every invariant I-1..I-15 carries exactly one of two statuses — ENFORCED (an executable check that raises on violation) or PENDING (the guarded feature does not exist yet). There is *no third state*: PENDING never means

“we chose not to check,” and no invariant may be absent or vacuously satisfied. When a feature lands, its invariant moves to ENFORCED *in the same change* or the auditor rejects the diff. Current state: **14 of 15 enforced**; only **I-10 (thin planner) is pending**, because the planner is not built. The checks are aggressively non-vacuous — each proves it *bites* against a planted mutant (a choosing render, an accumulator field, an η -KL tether, a web import, an EOC-forked settlement guard). Table 5 lists all fifteen.

Table 5: The 15 invariants and enforcement state (spec §14 + connector I-15).

	Invariant	Enforced
I-1	single tilt jack — no control path but the h-transform tilt	✓
I-2	gauge law — no coordinate crosses a track boundary	✓
I-3	no pressure accumulator / duplicate smoothing	✓
I-4	one F — no training loss distinct from F	✓
I-5	meters never in objective/gradient/settlement	✓
I-6	no external negatives; scramble family fixed in PREREG	✓
I-7	all interventions are clamped cells; no recovery modes	✓
I-8	streaming stability; halt on state growth (stationary input)	✓
I-9	run-time controls are tilt parameters only; F weights frozen	✓
I-10	planner stateless, external, thin	pending
I-11	rendering applies, never chooses	✓
I-12	provenance — every sample traces to (track, unit, transform)	✓
I-13	no browser/web tech in the runtime	✓
I-14	Hankel/holonomy are instruments; no fork of F ’s authority	✓
I-15	no premature aggregation / structure-deleting projection	✓

Builder/auditor adversarial pair. Implementation is done by an *ets-builder* agent and must be paired with a read-only *ets-auditor*; no gate runs and no merge happens without an auditor PASS. The auditor reports verdicts and never fixes code.

Prereg-commit-before-run, append-only registry. Every gate’s hypothesis, procedure, null, and kill condition is appended to `PREREG.md` and committed *before* the run; `REGISTRY.jsonl` is append-only. Corrections are new entries, never edits.

Operating slogan. *Walls are information, patches are sabotage, silent divergence is the breach class.* A wall (a failed gate, a non-identifiable scale, an unmeetable deadline) is surfaced and owned; the failure mode the discipline exists to catch is a *silent* divergence between what the spec defines and what the code does.

Case study — the fidelity breach (the discipline working)

The most instructive event in the build is a *fidelity breach caught before delivery* (registry `fidelity-breach-2026-07-`). An early implementation silently collapsed the unit-resolved coupling π to its marginal O — a structure-deleting conversion shim. The spec defines π as unit-resolved (§2/§5); the implementation aggregated without presenting the divergence. It was caught not by luck but by the *pre-registered exam*: the fixed scramble family correctly refused to separate (grid-shuffle held-out ≈ 0.35 , negative margin), the KILL condition fired, and — critically — no LAMBDA was emitted and the wall was surfaced pre-delivery. Remediation: (1) F re-posed on the full fiber; (2) a new invariant **I-15** with

two teeth — a term-input contract in `f.py` behaviorally verified against an O -preserving π rearrangement, and a referee-non-degeneracy check that the scored corpus-time feature *must* consume a unit-resolved fiber. The incident is logged `logged-permanent-no-relitigation`. This makes the paper *stronger*: the mechanism meant to catch an unfaithful implementation did, and left a durable guard behind.

6 Honest limitations — what is real vs. pending

The project’s identity forbids overselling. Table 6 is the honest state as of v1.

Table 6: Status ledger. Declared approximations and unpassed gates are reported, not rounded up.

Item	Status	Note
Ingestion $\rightarrow$ world $\rightarrow F \rightarrow$ writer $\rightarrow$ render	REAL	deterministic pure render (H-8 receipt)
Anchor self-sizing (eff. rank of traffic)	REAL	$M = 5$ on v0 corpus; G1 passed
Contrastive NCE fit of LAMBDA (rev-r1)	REAL	min held-out sep $0.95 \geq 0.90$; audit-reproduced bit-exactly
14/15 invariants enforced	REAL	only I-10 pending
Two-process engine+panel over OSC, MIDI CC learn	REAL	headless-graceful; $L = 2$ bars declared
G0 ingestion/reconstruction	pass	recon rel-L2 $\approx 1.16 \times 10^{-8}$
G1 anchor double dissociation	pass	diversity gap 2.04; SAME 2.38 < NULL 3.30 < DIVERSE 4.43; flat-in- N
G2 holonomy above calibrated null	did not pass	$\kappa_{\text{real}}/\kappa_{\text{null}} \approx 0.45$; reported as-is, no retune
REGION / CONTINUITY / NOVELTY / TEMPERATURE lanes	ARMED	derived σ_φ scales
DENSITY / GAUGE lanes	DISARMED	$\sigma_\varphi = 0$ at $u=0$; λ undefined, no tilt, surfaced
Greedy fiber realization	DECLARED APPROX	uncertified vs a joint fiber settlement; PASS-legal with declaration
Streaming temperature sampler	DECLARED APPROX	per-slot Laplace 2nd-order + orthant clip (~ 0.11 std bias, $\sim 13\%$ clip at $T_s=1$)
mass = $\sqrt{e[b]}$ settled-mass render	DECLARED APPROX	exact on schedule; adjacent-band crossover leakage ($\sim +0.15$ – 0.20 worst case)
I-10 planner	pending	closed-loop form thesis (G3–G6) untested
Mono	DECLARED V0	stereo restoration is real future scope, not a patch
External output master	OUTSIDE OBJECT	opt-in comp $\rightarrow$ R128 $\rightarrow$ limiter; pure render byte-identical by default
$u=0$ dynamic range / near-silence	OPEN FINDING	wide per-bar mass + global peak-normalize; no extrinsic AGC (structure-deleting class)

G2 in detail (the headline that did not land). G2 tests whether the frozen world carries *holonomy* (loop-defect curvature) above a residual-conditioned null — the dissociation that would support the conjecture that musical structure is real geometric curvature, at a demanding $20\times$ target. It **failed** (`g2_results.json`): real median curvature came in *below* the null ($\kappa_{\text{real}}/\kappa_{\text{null}} \approx 0.45$), dominance over null p95/p99 both false, `signal_present_above_null=false`. Real curvature *did* clear the solver floor ($\kappa_{\text{real}}/\kappa_{\text{floor}} \approx 2.0$), but not the residual-conditioned null. Per the pre-registered kill condition this is reported as-is, with no null-swap and no retune. Two consequences stated plainly: (i) the strong “curvature = structure” claim is *unsupported at v0*; (ii) the instrument survives as a read-only meter (`loop`) regardless. The pairwise-blindness and closed-loop *form* theses (G3–G6) depend on the planner, which is not built (I-10 pending), so they remain untested.

7 What ETS is, in one paragraph

ETS is an instrument whose “world” is a single Gibbs free energy fit so that real tracks are its equilibria, played by a causal writer that clamps its own frontier in real time and steered by an exact exponential tilt on gauge-invariant statistics whose knob scales are derived from equilibrium fluctuations. Its output is a settled tape of real source units, provenance-complete, not a synthesized sample. Its distinctive contribution is a *method*: one authority (F), two-state invariants with executable non-vacuous checks, an adversarial auditor, and an append-only registry where a caught fidelity breach and a failed headline gate are recorded as first-class results. The honest ledger — 14/15 invariants enforced, two lanes disarmed, one greedy realization, a failed G2, a pending planner — is not an apology; it is the artifact.

All figures are from the committed sources named inline (`ets-spec-v0.md`, `ets-connector-v0.md`, `REGISTRY.jsonl`) and the results artifacts `g0_results.json`, `g1_results.json`, `g2_results.json`, `training_results.json`, `latency_desktop.json`, and `ets/calibration/sigma_phi.json`. This document is post-v1 documentation; it modifies no source and affects no workflow.